Implicit State Changes in Procedural Text: Probing vs Fine-tuning

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Background and motivation

Language models are now used to plan actions, in robotics and in assistants that follow instructions. Such a system has to know where things end up even when the text never says so: that the fuel is gone once it has burned, or that the key is in the garage after the box it was in was carried there. If the model gets these implicit changes wrong, it plans from a wrong state. What I want to know is whether the information is missing from the model or only from its answer, since in the second case a classifier on the frozen model recovers it cheaply. Drożdż and Heilbron (2026) show that models from 410M parameters answer entity-tracking questions at human level, but asking a model does not show whether it represents the state internally, and earlier probing work looked at board games (Li et al., 2023) or at models fine-tuned on the domain (Li, Nye and Andreas, 2021).

Objective and research questions

The objective is to find out whether pretrained models represent state changes the text does not state, and whether fine-tuning adds that ability or only makes existing information easier to read out.

RQ1: How accurately can an entity's state change and location be classified from the frozen hidden states of a pretrained model?

RQ2: How far does accuracy drop when the entity is not mentioned in the sentence that changes its state, compared with the baselines?

RQ3: How does this depend on model size, and how large is the gap between a frozen probe and a fine-tuned model?

Main hypothesis

Pretrained models already represent implicit state changes, so probes on frozen hidden states will beat the baselines on implicit instances, and the gap to the fine-tuned model will shrink as models get larger. If the probes stay at baseline on implicit instances while the fine-tuned model is clearly better, I reject the hypothesis.

Data

I will use ProPara (Dalvi et al., 2018): 488 paragraphs about simple scientific processes, written by crowdworkers and annotated with 81,345 labels for where each participant is after each step. The grids were made by expert annotators and checked by a second one, locations by five workers with 0.67 average pairwise agreement, and human accuracy is about 80% against 34-45% for the paper's own models. The official 80/10/10 split is by process prompt, so test paragraphs are unseen processes. As a second, more controlled dataset I will use bAbI task 2 (Weston et al., 2016), where an object's location follows from the movements of the person carrying it: machine-generated, but an established benchmark rather than my own data.

Planned approach

The task is classification. One instance is one entity at one step, labelled from the annotation with its state change (CREATE, MOVE, DESTROY or NONE) and its location afterwards, chosen among the locations annotated in that paragraph plus "unknown" and "does not exist". Instances are split into explicit, where the entity is mentioned in the sentence, and implicit, where it is not. That split comes from the annotations and is what the study compares.

I will do the same classification in two ways: by fine-tuning, with a classification head on Pythia-160M, and by probing, with the model frozen and logistic regression trained on the hidden states of each layer, plus a control task that checks the probe is not learning the task by itself (Hewitt and Liang, 2019) and probes on the untrained model. Probes are run on Pythia 160M to 2.8B on ProPara and 160M to 1B on bAbI (Biderman et al., 2023), trained on the same data in the same order, so size is the only difference. The baselines are the majority class, keeping the entity's previous location, and the location mentioned most recently. I will report accuracy and macro-F1 separately for explicit and implicit instances, and the ROC-AUC of the predicted change, with bootstrap confidence intervals. If the frozen probe is close to the fine-tuned model, fine-tuning mostly surfaces what is already there, which is what Prakash et al. (2024) found for entity tracking.

Assumptions and risks

I take the annotations as correct, but an agreement of 0.67 puts the ceiling below 100%, so I will read a sample myself. ProPara and bAbI may both be in Pythia's training data (the Pile), which would flatter the results and which I cannot rule out. The main risk is that implicit instances are rare in ProPara; if so I will lean on bAbI task 2 or OpenPI (Tandon et al., 2020). I will include a short data statement (Bender and Friedman, 2018).

Expected outcome

I expect both methods to do well on explicit instances and to drop on implicit ones, with the probes above the baselines and better for larger models, and fine-tuning ahead by a shrinking margin. Probes at baseline would be just as useful a result, since the information would then have to come from fine-tuning.

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